Yuduo Jin

yuduojin@uvic.ca | linkedin

Education

University of Victoria Victoria, Canada
- PhD in Computer Science Jan 2024 - Jan 2028
- Supervisor: Prof. Brandon Haworth

Northeastern University Shenyang, China
- M.Eng. in Computer Technology Sep 2020 - Jul 2023

Liaoning Normal University Dalian, China
- B.Eng. in Digital Media Technology Sep 2016 - Jul 2020

Research Experience

Language-Driven Multi-Agent Model for Emergent Crowd Dynamics Apr 2026 - Present
- Developed an LLM-driven multi-agent crowd simulation framework to enable dialogue-driven dynamic interactions, bridging **agent reasoning**, **trajectory planning**, and **motion matching** for realistic 3D human motion synthesis from optimized trajectories.
- Built a motion retargeting and rendering pipeline using Mixamo characters in Blender, creating indoor and outdoor environments; generated **long-term crowd animations** and evaluated the results for motion realism and social behavior consistency.

City-Level GPS Trajectory Generation Using Tile-Conditioned Diffusion Model. (Under Review) Jan 2026 - Present
- Developed a human mobility trajectory-processing pipeline for **5 cities**, covering data cleaning, trajectory resampling, spatio-temporal correlation modeling, and quality control. Established a unified training framework to provide high-quality data foundations for city-level trajectory generation.
- Proposed a **tile-conditioned diffusion model** that incorporates map tiles and start locations as spatial conditions for **city-level GPS trajectory synthesis**, generating realistic and diverse trajectories with strong generalization while preserving authentic mobility dynamics and enabling privacy-aware data sharing for downstream urban computing applications.

Temporal Convolutional Network-Based Motion Refinement for Enhanced Human Motion Prediction May 2025 - Present
- Addressed motion artifacts caused by accumulated errors in long-term human motion prediction by proposing a Temporal Convolutional Network (TCN)-based motion refiner to improve temporal consistency of predicted sequences. Achieved average reductions of **2.81% and 3.49% in mean joint position error (MPJPE)** over MLP and diffusion-based baselines, respectively.
- **Constructed a paired noisy-clean motion dataset** for supervised training and developed automated data processing and visualization pipelines. **Led a 4-member team** in data preparation, neural network development, experimental evaluation, and result analysis to support iterative model optimization.

Back to Basics: Motion Representation Matters for Human Motion Generation Using Diffusion Model (arXiv:2512.04499. Under Review) Jan 2025 – Present
- Presented a comprehensive empirical analysis of motion representations for Motion Diffusion Models (MDM), establishing a unified evaluation framework across **six widely used representations**. Quantitatively analyzed the trade-offs among representation choices, generation fidelity, and computational efficiency.
- Benchmarked the proposed evaluation framework on HumanAct12, 100STYLE, and HumanML3D, revealing that compact position-based representations with v-parameterization objectives provide improved generation quality and training efficiency. The study offers actionable **guidelines for designing motion representations and training strategies** in human motion generation.

Academic Service

Reviewer, Annual Conference of the European Association for Computer Graphics (EG)
Reviewer, ACM SIGGRAPH conference on Motion, Interaction and Games (MIG)
Reviewer, Computer Animation and Virtual Worlds (CAVW)

Teaching Experience

Teaching assistant, CSC 305 Introduction to Computer Graphics (Sep 2026)
Teaching assistant, CSC 305 Introduction to Computer Graphics (May 2025)
Teaching assistant, CSC 305 Introduction to Computer Graphics (Jan 2025)
Teaching assistant, CSC 473/573 Fundamentals of Computer Animation (Sep 2024)

Others

Research Interests: Human motion synthesis, motion capture, human mobility, character animation, crowd simulation, generative models
Volunteer: Student volunteer, SIGGRAPH 2025 & 2026
Programming: Python, C++, C, Java, SQL
Technologies: PyTorch, TensorFlow, Git, Blender, Unity, 3ds Max (3D modeling), MotionBuilder, MATLAB, Adobe Photoshop (UI/Graphics design), Illustrator, Premiere
Relevant Courses: Fundamentals of Computer Animation, Data Mining, Computer Graphics, Digital Image Processing, Computer Vision, VR Technology, Artificial Intelligence, HCI, Basis of Fine Art